

Breast Cancer Diagnosis using Machine Learning

Formative Assessment-02 · Data Analysis Case Study & Streamlit Deployment

Advanced Data Science [MCA33PE17] · MCA · SYMCA · Academic Year 2026-2027 · Semester I

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Live Streamlit app: <https://DS-Breast-Cancer.streamlit.app>

1. Problem and dataset

Goal: predict whether a breast tumour is **malignant** or **benign** from measurements of a digitised fine-needle-aspirate image, and compare four classification algorithms. **Dataset:** *Breast Cancer Wisconsin (Diagnostic)* – UCI Machine Learning Repository (Wolberg, Street & Mangasarian, 1995), 569 samples × 30 numeric features (10 nucleus measurements, each as mean, standard error and worst value); 357 benign (62.7 %) and 212 malignant (37.3 %).

Why this dataset: a real, high-impact screening problem where the *type* of error matters; clean numeric data so the focus stays on modelling; moderate class imbalance that justifies precision/recall/F1 over accuracy; strongly correlated features that make scaling and regularisation meaningful; and small enough that all four models can be tuned and cross-validated fairly. Metrics are reported for the **malignant** class, because a missed cancer (false negative) is the costly error.

2. Exploratory data analysis – key insights

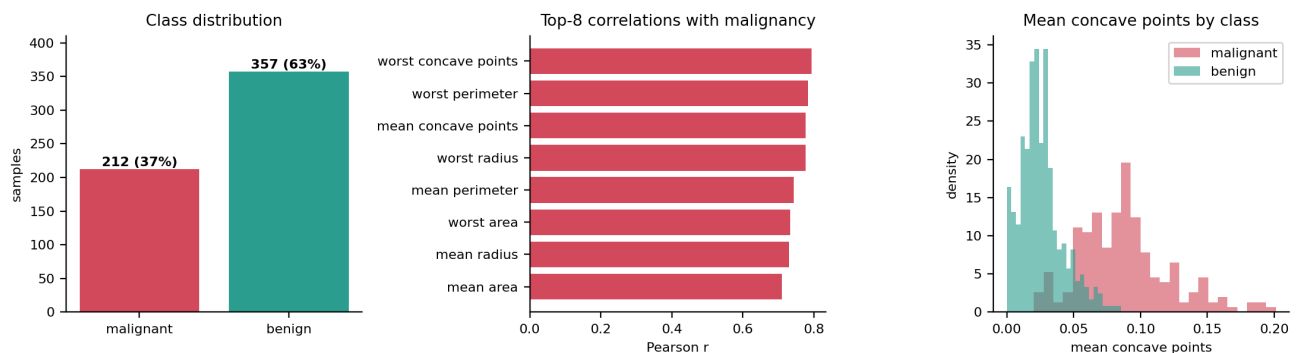


Figure 1 – class balance, strongest correlations with malignancy, and class-wise distribution of one key feature.

- **Data quality is high:** no missing values, no duplicates, all 30 features numeric – nothing to impute or encode.
- **Size and shape drive malignancy:** median mean area 932 vs 458 (2.0×), mean concave points 0.086 vs 0.023 (3.7×); worst concave points, worst perimeter and mean concave points correlate ≈ 0.78 – 0.79 with malignancy. Five features (e.g. fractal dimension, texture error) carry almost no linear signal ($|r| < 0.1$).
- **Strong multicollinearity:** 21 of 435 feature pairs have $|r| > 0.9$ (radius $\leftrightarrow$ perimeter $\leftrightarrow$ area), so regularised or tree-based models are appropriate.
- **Outliers are genuine:** 608 IQR-flagged values (171 of 569 rows), mostly right-skewed “error” features; they are large tumours, so they were kept.
- **Different scales** (mean area up to 2501 vs smoothness ≈ 0.1) $\rightarrow$ scaling is required for Logistic Regression and SVM.

3. Preprocessing

- **Missing values / encoding:** none present; a median *SimpleImputer* is still the first pipeline step as a safeguard. No categorical features; the target is already label-encoded (0 = malignant, 1 = benign).
- **Scaling:** *StandardScaler* for Logistic Regression and SVM only (trees are scale-invariant), placed *inside* a *Pipeline* so it is fitted on training data only (no leakage).
- **Split:** stratified 80 % train (455) / 20 % test (114), random_state = 42; both sets keep ≈ 37 % malignant.

4. Models, cross-validation and tuning

Four algorithms were implemented with scikit-learn: **Logistic Regression, Decision Tree, Random Forest and SVM (RBF)**. Each was first checked with **stratified 5-fold cross-validation** on the training data (mean $\pm$ std shows stability; the untuned Decision Tree scores 100 % on training folds but only ≈ 92 % in CV, i.e. it over-fits). **GridSearchCV** (same 5 folds, scoring = F1 of the malignant class) was then applied

to **all four** models; the test set was never used to choose parameters.

Model	Best parameters (GridSearchCV)	Combos	Baseline CV F1	Tuned CV F1
Logistic Regression	{'C': 0.1, 'class_weight': None}	10	0.9703	0.9760
Decision Tree	{'criterion': 'entropy', 'max_depth': 4, 'min_samples_leaf': 5}	56	0.8919	0.9174
Random Forest	{'max_depth': None, 'min_samples_split': 2, 'n_estimators': 300}	36	0.9506	0.9533
SVM	{'C': 10, 'gamma': 0.01}	16	0.9583	0.9668

Table 1 – GridSearchCV results: tuning raised cross-validated F1 for every model, most for the Decision Tree.

5. Results and model comparison (unseen test set, malignant = positive class)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Missed malignant	False alarms	Baseline F1
SVM	0.982	0.976	0.976	0.976	0.998	1	1	0.976
Logistic Regression	0.974	0.976	0.952	0.964	0.996	2	1	0.976
Random Forest	0.947	0.929	0.929	0.929	0.994	3	3	0.940
Decision Tree	0.877	0.780	0.929	0.848	0.959	3	11	0.886

Table 2 – tuned models on the 114 test records (best model highlighted); last column shows the untuned baseline F1.

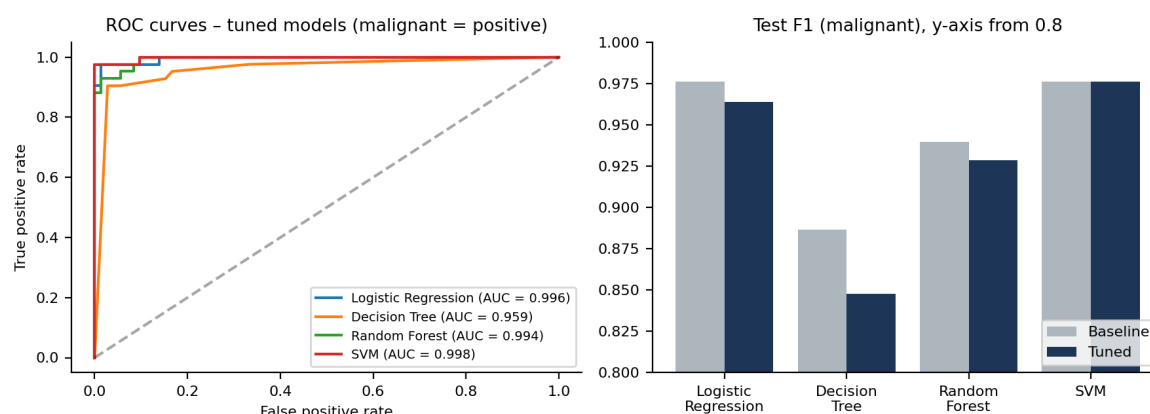


Figure 2 – ROC curves of the tuned models and baseline-vs-tuned F1.

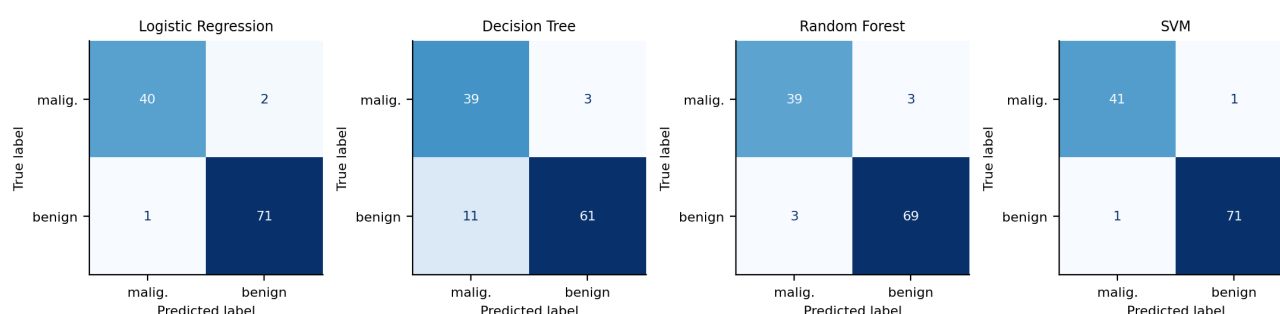


Figure 3 – confusion matrices (rows = actual, columns = predicted).

6. Insights and conclusions

- **Best model: SVM** – F1 0.976, ROC-AUC 0.998, only 1 of 42 malignant tumours missed. Logistic Regression is a very close second and is the more explainable choice.
- **Ranking SVM ≈ Logistic Regression > Random Forest > Decision Tree** is the same in cross-validation and on the test set. The classes are almost linearly separable in standardised space, which suits smooth boundaries better than axis-aligned tree splits.
- **Ensembling fixes over-fitting:** Random Forest beats a single Decision Tree by ≈ 5 F1 points.
- **Tuning gains are real in cross-validation but small on the test set:** with only 114 test records (42 malignant) one prediction moves recall by 2.4 points, so models were selected on cross-validation and the test set was used only for reporting.
- **What drives predictions:** six of the top-10 features are shared by Random Forest importances and Logistic Regression coefficients – all size / concavity measures (worst radius / perimeter / area, concave

points), matching the EDA. They differ on texture: Logistic Regression ranks *worst texture* first (weakly correlated with size, $r \approx 0.35$, so it adds complementary evidence), Random Forest ranks it 12th.

- **Limitations:** single dataset, small test set, no external validation. This is an educational model, **not a diagnostic tool**.

7. Streamlit application (deployment)

app.py loads the four tuned models saved with *joblib* and provides: (1) an input form with 30 grouped sliders (mean / standard error / worst) inside a **Predict** button flow, plus one-click examples – typical benign, typical malignant, random real test record; (2) results showing **every model's prediction and malignant probability** and a consensus vote; (3) a **Model comparison** tab (metrics, confusion matrices, tuning results computed live on the test split); (4) a **Dataset** tab (class balance, feature importance, data preview). Run locally with *streamlit run app.py*.

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Educational demo – predictions are not a medical diagnosis.